

# Digital Design (SET 111)

## L09

# Boolean Algebra and Logic Simplification

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- ☐ Exercise
- ☐ “Don’t Care” Conditions



## ☐ Exercise

## ☐ “Don’t Care” Conditions

# Exercise



□ Ex.1 Use a Karnaugh map to minimize the following standard SOP expression:

$$F = A \cdot \overline{B} \cdot C + \overline{A} \cdot B \cdot C + \overline{A} \cdot \overline{B} \cdot C + \overline{A} \cdot \overline{B} \cdot \overline{C} + A \cdot \overline{B} \cdot \overline{C}$$

Ans

$$F = \overline{B} + \overline{A} \cdot C$$

|   |   | B  |    |    |    |
|---|---|----|----|----|----|
|   |   | 00 | 01 | 11 | 10 |
| A | 0 | 1  | 1  | 1  | 0  |
|   | 1 | 1  | 1  | 0  | 0  |

# “Don’t Care” Conditions (1/2)



- ❑ Sometimes a situation arises in which some **input** variable combinations are **not allowed**.
- ❑ That is, for these “**don’t care**” terms either a **1** or a **0** may be assigned to the output; it really does **not matter** since they will never occur.
- ❑ The “don’t care” terms can be used to advantage on the **Karnaugh** map.
- ❑ For each “don’t care” term, an **X** is placed in the cell. When grouping the 1s, the Xs can be treated as 1s to make a larger grouping or as 0s if they cannot be used to advantage.

# “Don’t Care” Conditions (2/2)



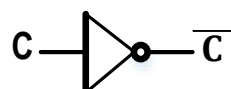
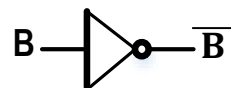
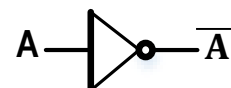
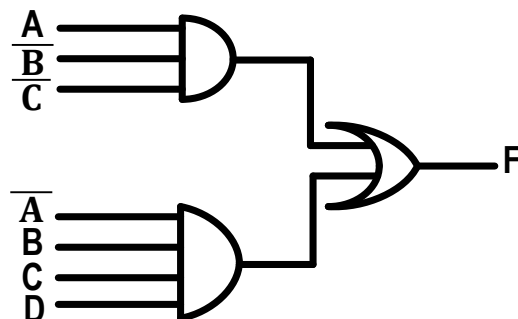
## □ Example.

### Truth Table

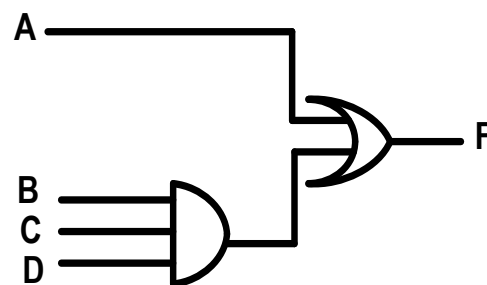
| Input |   |   |   | Output |
|-------|---|---|---|--------|
| A     | B | C | D | F      |
| 0     | 0 | 0 | 0 | 0      |
| 0     | 0 | 0 | 1 | 0      |
| 0     | 0 | 1 | 0 | 0      |
| 0     | 0 | 1 | 1 | 0      |
| 0     | 1 | 0 | 0 | 0      |
| 0     | 1 | 0 | 1 | 0      |
| 0     | 1 | 1 | 0 | 0      |
| 0     | 1 | 1 | 1 | 1      |
| 1     | 0 | 0 | 0 | 1      |
| 1     | 0 | 0 | 1 | 1      |
| 1     | 0 | 1 | 0 | X      |
| 1     | 0 | 1 | 1 | X      |
| 1     | 1 | 0 | 0 | X      |
| 1     | 1 | 0 | 1 | X      |
| 1     | 1 | 1 | 0 | X      |
| 1     | 1 | 1 | 1 | X      |

Don't Care

□ Without don't care  
 $F = A \cdot \overline{B} \cdot \overline{C} + \overline{A} \cdot B \cdot C \cdot D$



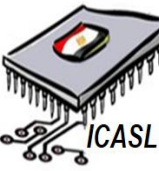
□ With don't care  
 $F = A + B \cdot C \cdot D$



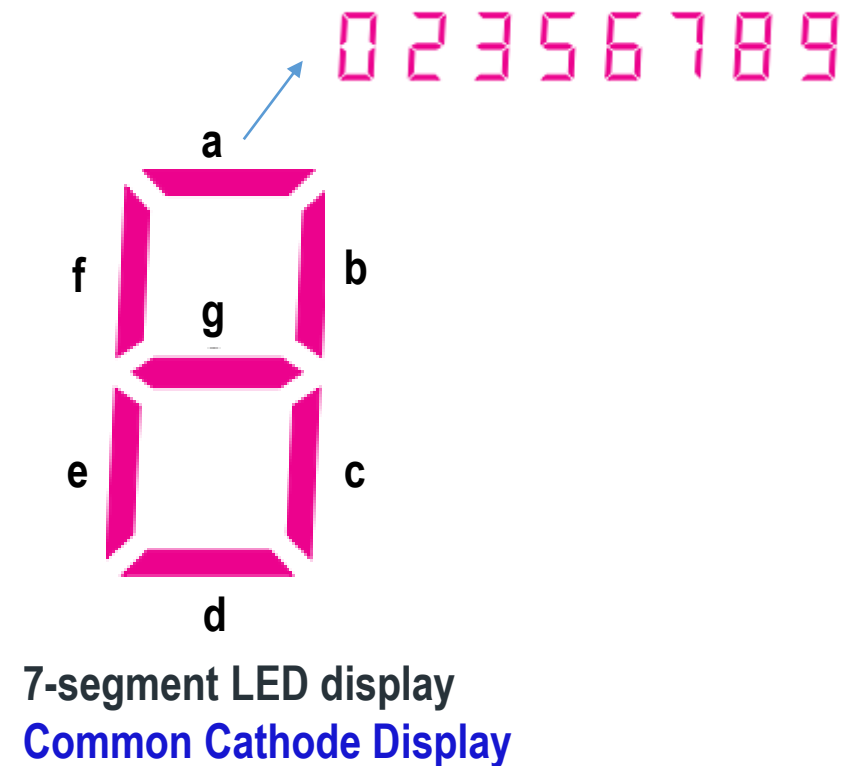
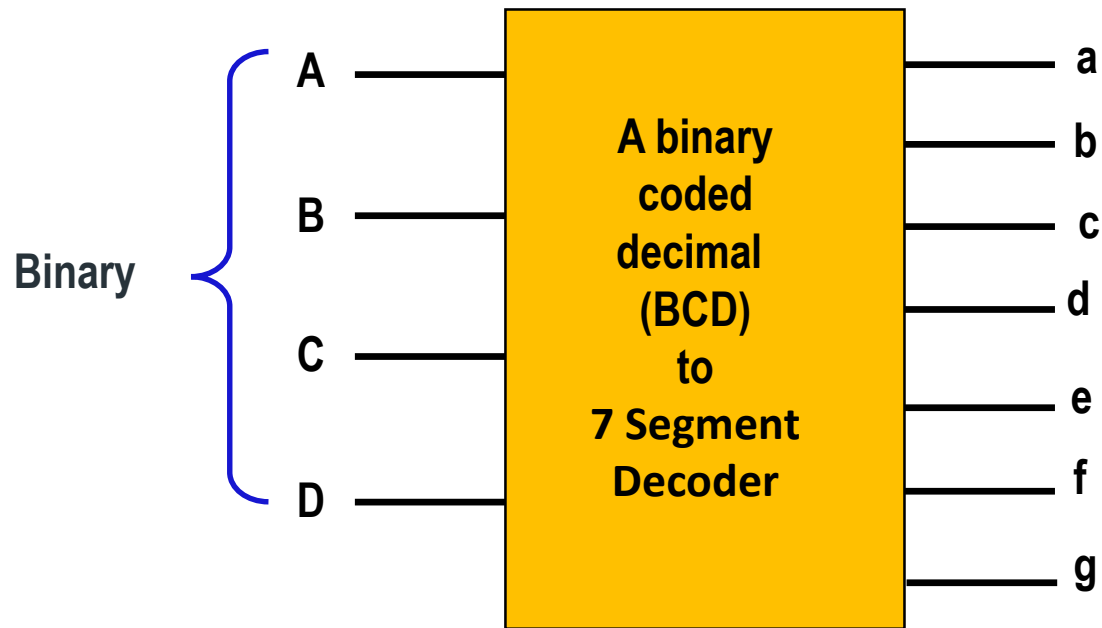
|   |    | C  |    |    |    |
|---|----|----|----|----|----|
|   |    | 00 | 01 | 11 | 10 |
| A | 00 | 0  | 0  | 0  | 0  |
|   | 01 | 0  | 0  | 1  | 0  |
|   | 11 | x  | x  | x  | x  |
|   | 10 | 1  | 1  | x  | x  |

|   |    | D  |    |    |    |
|---|----|----|----|----|----|
|   |    | 00 | 01 | 11 | 10 |
| A | 00 | 0  | 0  | 0  | 0  |
|   | 01 | 0  | 0  | 1  | 0  |
|   | 11 | x  | x  | x  | x  |
|   | 10 | 1  | 1  | x  | x  |

# Exercise (1/15)



- In a 7-segment display, each of the seven segments is activated for various digits. For example, segment **a** is activated for the digits 0, 2, 3, 5, 6, 7, 8, and 9, as shown. Since each digit can be represented by a BCD code, **derive** an SOP expression for segment **a** using the variables ABCD and then minimize the expression using K-map.

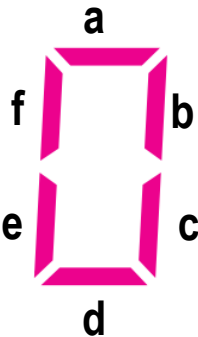


# Exercise (2/15)



## Truth Table

| Input |   |   |   | Output |   |   |   |   |   |   |
|-------|---|---|---|--------|---|---|---|---|---|---|
| A     | B | C | D | a      | b | c | d | e | f | g |
| 0     | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0     | 0 | 0 | 1 |        |   |   |   |   |   |   |
| 0     | 0 | 1 | 0 |        |   |   |   |   |   |   |
| 0     | 0 | 1 | 1 |        |   |   |   |   |   |   |
| 0     | 1 | 0 | 0 |        |   |   |   |   |   |   |
| 0     | 1 | 0 | 1 |        |   |   |   |   |   |   |
| 0     | 1 | 1 | 0 |        |   |   |   |   |   |   |
| 0     | 1 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 0 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 1 |        |   |   |   |   |   |   |

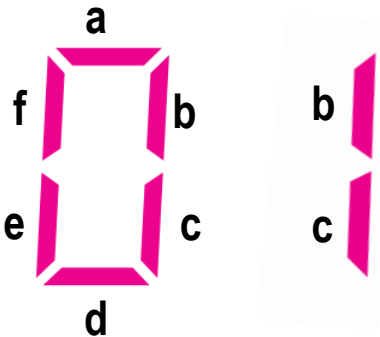


# Exercise (3/15)



## Truth Table

| Input |   |   |   | Output |   |   |   |   |   |   |
|-------|---|---|---|--------|---|---|---|---|---|---|
| A     | B | C | D | a      | b | c | d | e | f | g |
| 0     | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0     | 0 | 0 | 1 | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0     | 0 | 1 | 0 |        |   |   |   |   |   |   |
| 0     | 0 | 1 | 1 |        |   |   |   |   |   |   |
| 0     | 1 | 0 | 0 |        |   |   |   |   |   |   |
| 0     | 1 | 0 | 1 |        |   |   |   |   |   |   |
| 0     | 1 | 1 | 0 |        |   |   |   |   |   |   |
| 0     | 1 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 0 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 1 |        |   |   |   |   |   |   |

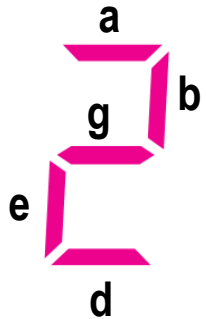
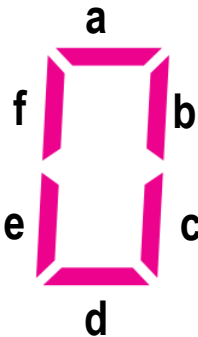


# Exercise (4/15)



## Truth Table

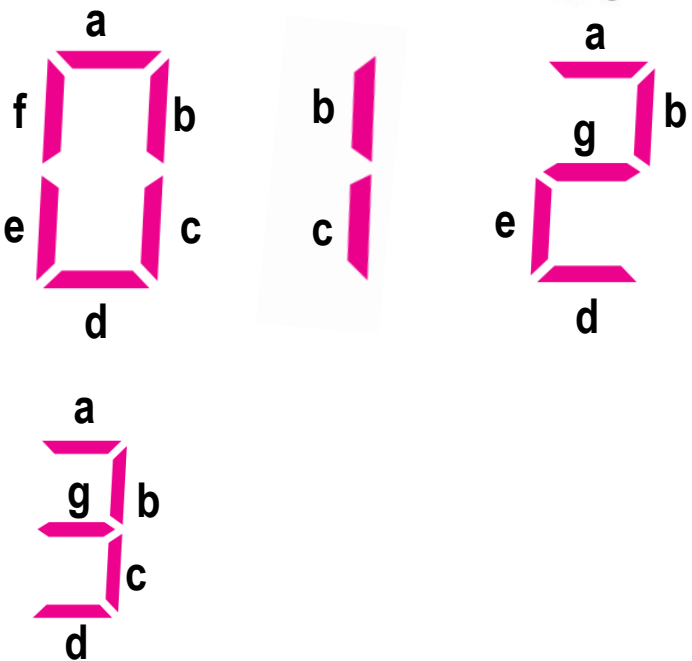
| Input |   |   |   | Output |   |   |   |   |   |   |
|-------|---|---|---|--------|---|---|---|---|---|---|
| A     | B | C | D | a      | b | c | d | e | f | g |
| 0     | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0     | 0 | 0 | 1 | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0     | 0 | 1 | 0 | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0     | 0 | 1 | 1 |        |   |   |   |   |   |   |
| 0     | 1 | 0 | 0 |        |   |   |   |   |   |   |
| 0     | 1 | 0 | 1 |        |   |   |   |   |   |   |
| 0     | 1 | 1 | 0 |        |   |   |   |   |   |   |
| 0     | 1 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 0 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 1 |        |   |   |   |   |   |   |



# Exercise (5/15)

## Truth Table

| Input |   |   |   | Output |   |   |   |   |   |   |
|-------|---|---|---|--------|---|---|---|---|---|---|
| A     | B | C | D | a      | b | c | d | e | f | g |
| 0     | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0     | 0 | 0 | 1 | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0     | 0 | 1 | 0 | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0     | 0 | 1 | 1 | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 0     | 1 | 0 | 0 |        |   |   |   |   |   |   |
| 0     | 1 | 0 | 1 |        |   |   |   |   |   |   |
| 0     | 1 | 1 | 0 |        |   |   |   |   |   |   |
| 0     | 1 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 0 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 1 |        |   |   |   |   |   |   |

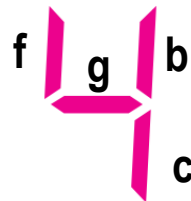
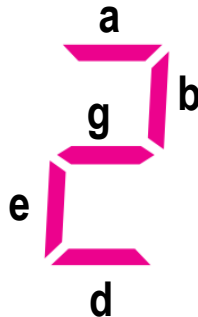
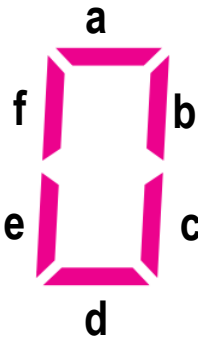


# Exercise (6/15)



## Truth Table

| Input |   |   |   | Output |   |   |   |   |   |   |
|-------|---|---|---|--------|---|---|---|---|---|---|
| A     | B | C | D | a      | b | c | d | e | f | g |
| 0     | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0     | 0 | 0 | 1 | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0     | 0 | 1 | 0 | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0     | 0 | 1 | 1 | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 0     | 1 | 0 | 0 | 0      | 1 | 1 | 0 | 0 | 1 | 1 |
| 0     | 1 | 0 | 1 |        |   |   |   |   |   |   |
| 0     | 1 | 1 | 0 |        |   |   |   |   |   |   |
| 0     | 1 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 0 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 1 |        |   |   |   |   |   |   |

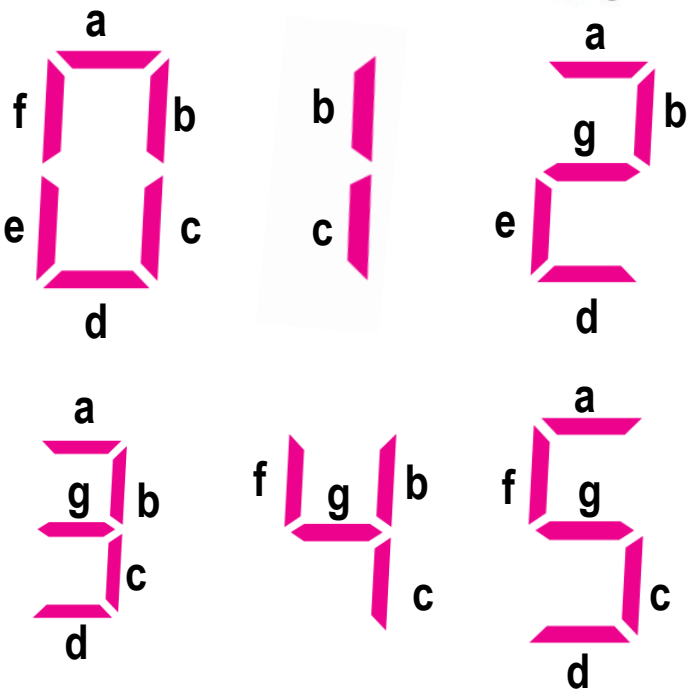


# Exercise (7/15)



## Truth Table

| Input |   |   |   | Output |   |   |   |   |   |   |
|-------|---|---|---|--------|---|---|---|---|---|---|
| A     | B | C | D | a      | b | c | d | e | f | g |
| 0     | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0     | 0 | 0 | 1 | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0     | 0 | 1 | 0 | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0     | 0 | 1 | 1 | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 0     | 1 | 0 | 0 | 0      | 1 | 1 | 0 | 0 | 1 | 1 |
| 0     | 1 | 0 | 1 | 1      | 0 | 1 | 1 | 0 | 1 | 1 |
| 0     | 1 | 1 | 0 |        |   |   |   |   |   |   |
| 0     | 1 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 0 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 1 |        |   |   |   |   |   |   |

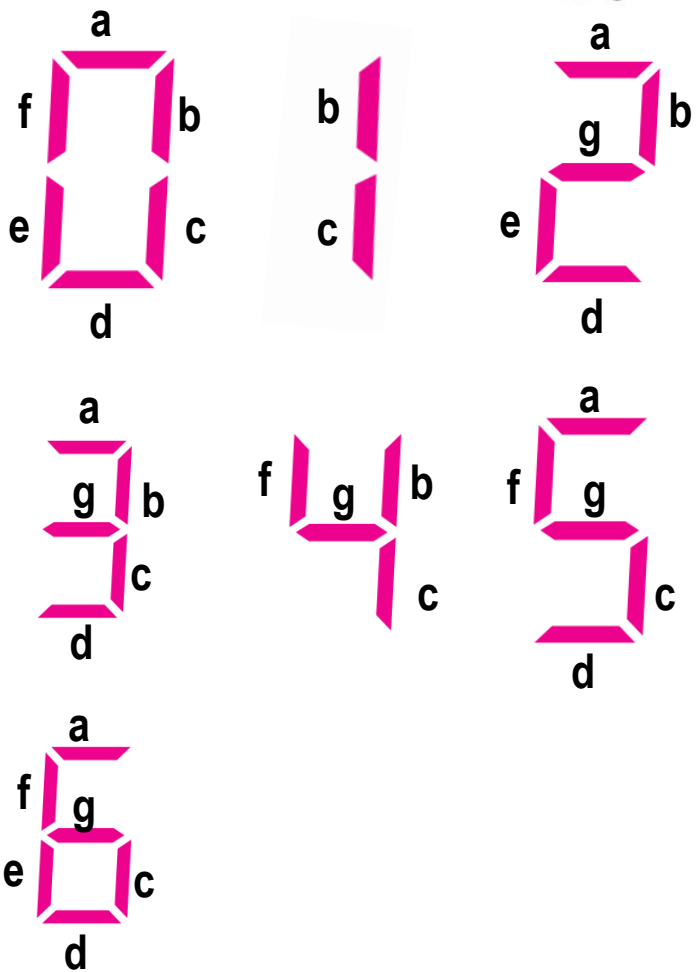


# Exercise (8/15)



## Truth Table

| Input |   |   |   | Output |   |   |   |   |   |   |
|-------|---|---|---|--------|---|---|---|---|---|---|
| A     | B | C | D | a      | b | c | d | e | f | g |
| 0     | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0     | 0 | 0 | 1 | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0     | 0 | 1 | 0 | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0     | 0 | 1 | 1 | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 0     | 1 | 0 | 0 | 0      | 1 | 1 | 0 | 0 | 1 | 1 |
| 0     | 1 | 0 | 1 | 1      | 0 | 1 | 1 | 0 | 1 | 1 |
| 0     | 1 | 1 | 0 | 1      | 0 | 1 | 1 | 1 | 1 | 1 |
| 0     | 1 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 0 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 1 |        |   |   |   |   |   |   |

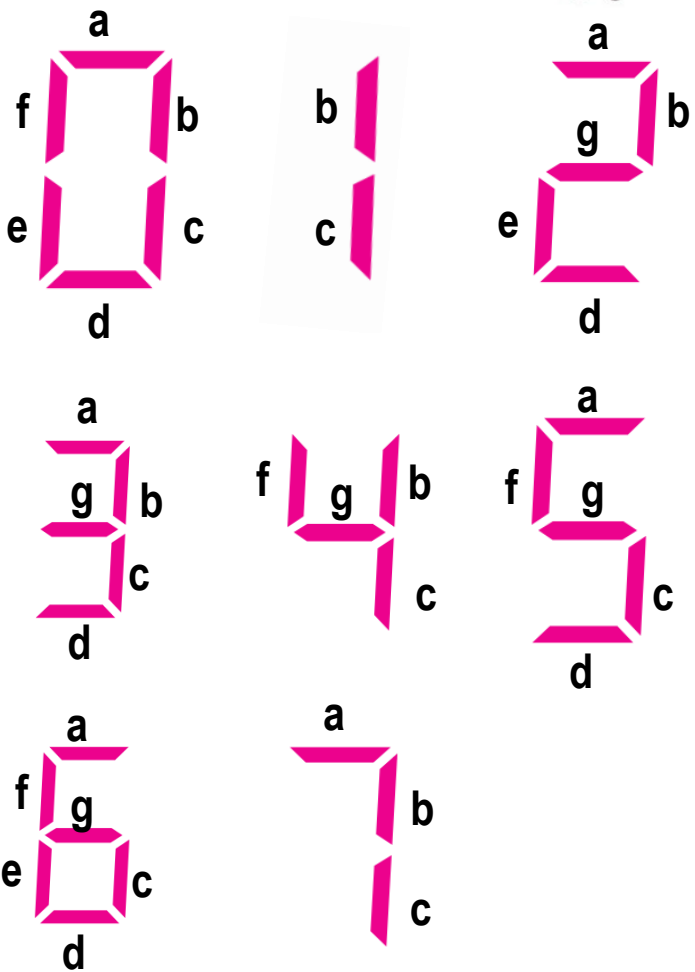


# Exercise (9/15)



## Truth Table

| Input |   |   |   | Output |   |   |   |   |   |   |
|-------|---|---|---|--------|---|---|---|---|---|---|
| A     | B | C | D | a      | b | c | d | e | f | g |
| 0     | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0     | 0 | 0 | 1 | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0     | 0 | 1 | 0 | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0     | 0 | 1 | 1 | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 0     | 1 | 0 | 0 | 0      | 1 | 1 | 0 | 0 | 1 | 1 |
| 0     | 1 | 0 | 1 | 1      | 0 | 1 | 1 | 0 | 1 | 1 |
| 0     | 1 | 1 | 0 | 1      | 0 | 1 | 1 | 1 | 1 | 1 |
| 0     | 1 | 1 | 1 | 1      | 1 | 1 | 0 | 0 | 0 | 0 |
| 1     | 0 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 1 |        |   |   |   |   |   |   |

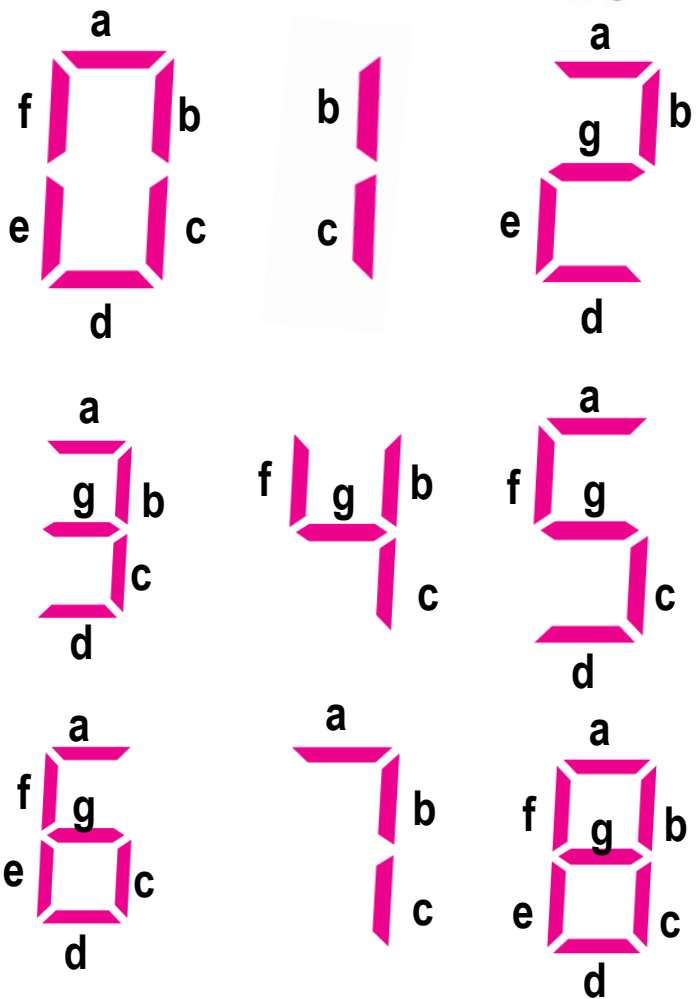


# Exercise (10/15)



## Truth Table

| Input |   |   |   | Output |   |   |   |   |   |   |
|-------|---|---|---|--------|---|---|---|---|---|---|
| A     | B | C | D | a      | b | c | d | e | f | g |
| 0     | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0     | 0 | 0 | 1 | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0     | 0 | 1 | 0 | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0     | 0 | 1 | 1 | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 0     | 1 | 0 | 0 | 0      | 1 | 1 | 0 | 0 | 1 | 1 |
| 0     | 1 | 0 | 1 | 1      | 0 | 1 | 1 | 0 | 1 | 1 |
| 0     | 1 | 1 | 0 | 1      | 0 | 1 | 1 | 1 | 1 | 1 |
| 0     | 1 | 1 | 1 | 1      | 1 | 1 | 0 | 0 | 0 | 0 |
| 1     | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 1 |
| 1     | 0 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 1 |        |   |   |   |   |   |   |

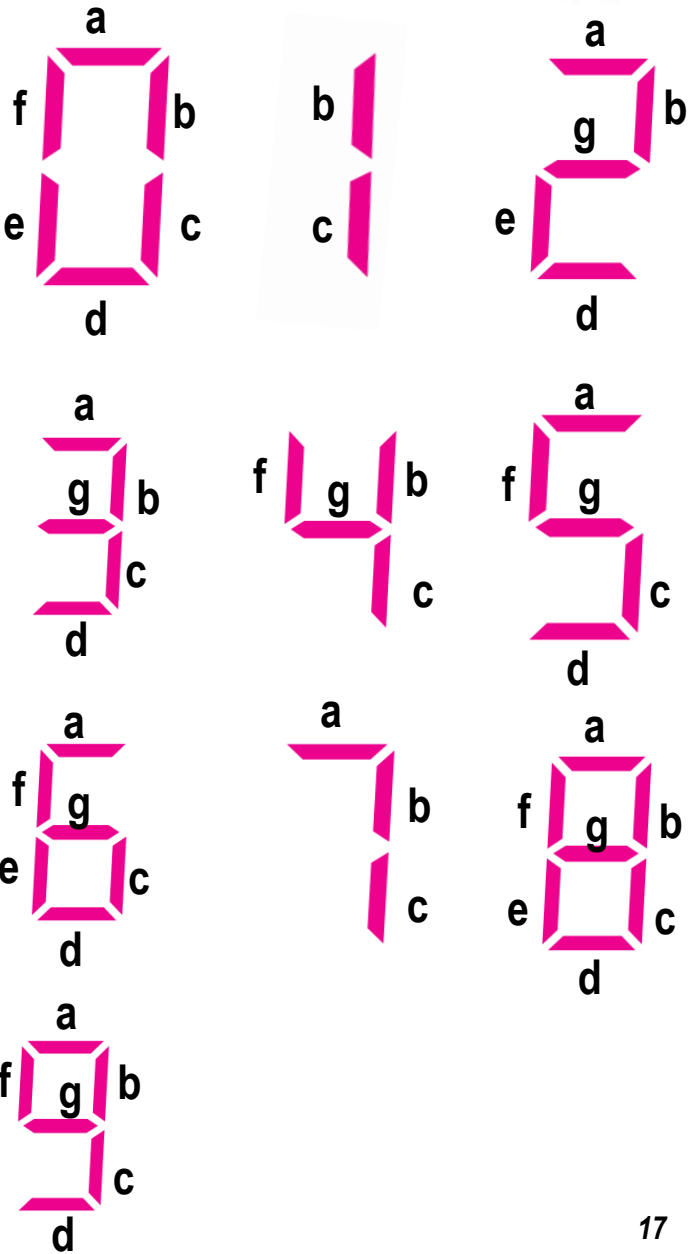


# Exercise (11/15)



## Truth Table

| Input |   |   |   | Output |   |   |   |   |   |   |
|-------|---|---|---|--------|---|---|---|---|---|---|
| A     | B | C | D | a      | b | c | d | e | f | g |
| 0     | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0     | 0 | 0 | 1 | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0     | 0 | 1 | 0 | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0     | 0 | 1 | 1 | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 0     | 1 | 0 | 0 | 0      | 1 | 1 | 0 | 0 | 1 | 1 |
| 0     | 1 | 0 | 1 | 1      | 0 | 1 | 1 | 0 | 1 | 1 |
| 0     | 1 | 1 | 0 | 1      | 0 | 1 | 1 | 1 | 1 | 1 |
| 0     | 1 | 1 | 1 | 1      | 1 | 1 | 0 | 0 | 0 | 0 |
| 1     | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 1 |
| 1     | 0 | 0 | 1 | 1      | 1 | 1 | 1 | 0 | 1 | 1 |
| 1     | 0 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 0 | 1 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 0 | 1 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 0 |        |   |   |   |   |   |   |
| 1     | 1 | 1 | 1 |        |   |   |   |   |   |   |

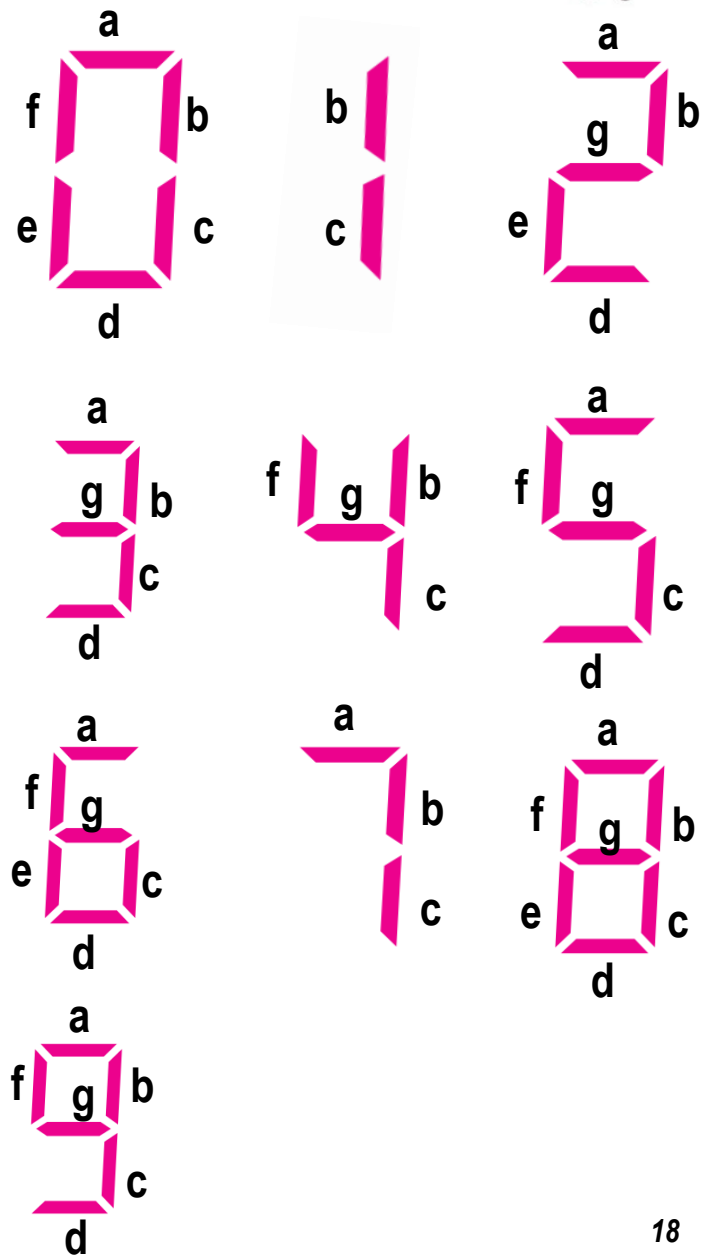


# Exercise (12/15)



## Truth Table

| Input |   |   |   | Output |   |   |   |   |   |   |
|-------|---|---|---|--------|---|---|---|---|---|---|
| A     | B | C | D | a      | b | c | d | e | f | g |
| 0     | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 0     | 0 | 0 | 1 | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 0     | 0 | 1 | 0 | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 0     | 0 | 1 | 1 | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 0     | 1 | 0 | 0 | 0      | 1 | 1 | 0 | 0 | 1 | 1 |
| 0     | 1 | 0 | 1 | 1      | 0 | 1 | 1 | 0 | 1 | 1 |
| 0     | 1 | 1 | 0 | 1      | 0 | 1 | 1 | 1 | 1 | 1 |
| 0     | 1 | 1 | 1 | 1      | 1 | 1 | 0 | 0 | 0 | 0 |
| 1     | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 1 |
| 1     | 0 | 0 | 1 | 1      | 1 | 1 | 1 | 0 | 1 | 1 |
| 1     | 0 | 1 | 0 | x      | x | x | x | x | x | x |
| 1     | 0 | 1 | 1 | x      | x | x | x | x | x | x |
| 1     | 1 | 0 | 0 | x      | x | x | x | x | x | x |
| 1     | 1 | 0 | 1 | x      | x | x | x | x | x | x |
| 1     | 1 | 1 | 0 | x      | x | x | x | x | x | x |
| 1     | 1 | 1 | 1 | x      | x | x | x | x | x | x |



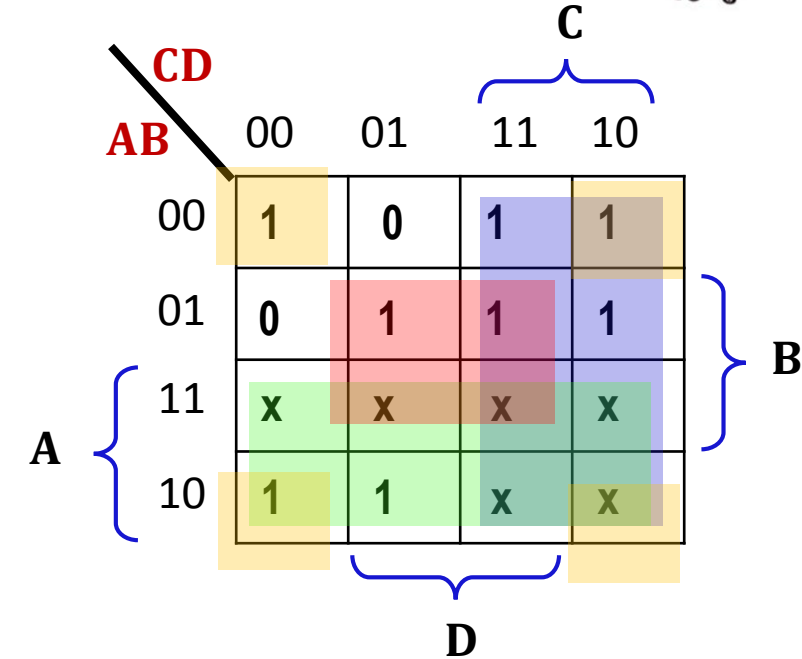
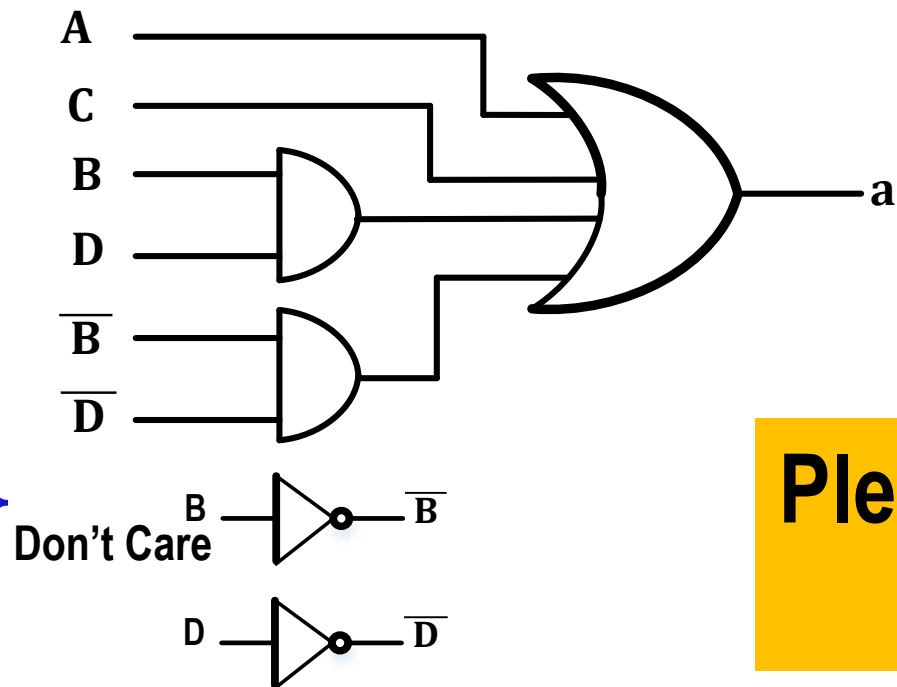
# Exercise (13/15)

## Truth Table

| Input |   |   |   | Output |
|-------|---|---|---|--------|
| A     | B | C | D | a      |
| 0     | 0 | 0 | 0 | 1      |
| 0     | 0 | 0 | 1 | 0      |
| 0     | 0 | 1 | 0 | 1      |
| 0     | 0 | 1 | 1 | 1      |
| 0     | 1 | 0 | 0 | 0      |
| 0     | 1 | 0 | 1 | 1      |
| 0     | 1 | 1 | 0 | 1      |
| 0     | 1 | 1 | 1 | 1      |
| 1     | 0 | 0 | 0 | 1      |
| 1     | 0 | 0 | 1 | 1      |
| 1     | 0 | 1 | 0 | x      |
| 1     | 0 | 1 | 1 | x      |
| 1     | 1 | 0 | 0 | x      |
| 1     | 1 | 0 | 1 | x      |
| 1     | 1 | 1 | 0 | x      |
| 1     | 1 | 1 | 1 | x      |

□ With don't care

$$a = A + C + B.D + \overline{B} . \overline{D}$$



Please, complete for  
b, c, d, e, f, g

# Exercise (14/15)

## Truth Table

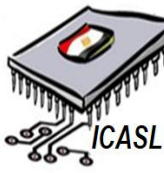
| Input |   |   |   | Output         |   |
|-------|---|---|---|----------------|---|
| A     | B | C | D | $\checkmark a$ | b |
| 0     | 0 | 0 | 0 | 1              | 1 |
| 0     | 0 | 0 | 1 | 0              | 1 |
| 0     | 0 | 1 | 0 | 1              | 1 |
| 0     | 0 | 1 | 1 | 1              | 1 |
| 0     | 1 | 0 | 0 | 0              | 1 |
| 0     | 1 | 0 | 1 | 1              | 0 |
| 0     | 1 | 1 | 0 | 1              | 0 |
| 0     | 1 | 1 | 1 | 1              | 1 |
| 1     | 0 | 0 | 0 | 1              | 1 |
| 1     | 0 | 0 | 1 | 1              | 1 |
| 1     | 0 | 1 | 0 | x              | x |
| 1     | 0 | 1 | 1 | x              | x |
| 1     | 1 | 0 | 0 | x              | x |
| 1     | 1 | 0 | 1 | x              | x |
| 1     | 1 | 1 | 0 | x              | x |
| 1     | 1 | 1 | 1 | x              | x |

|   |    | C  |    |    |    |
|---|----|----|----|----|----|
|   |    | 00 | 01 | 11 | 10 |
| A | 00 | 1  | 1  | 1  | 1  |
|   | 01 | 1  | 0  | 1  | 0  |
|   | 11 | x  | x  | x  | x  |
|   | 10 | 1  | 1  | x  | x  |

□ With don't care

$$b = \overline{B} + \overline{C} \cdot \overline{D} + C \cdot D$$

Please, complete for  
b, c, d, e, f, g



# Exercise (15/15)



## Truth Table

□ With don't care

$$a = A + C + B.D + \overline{B}. \overline{D}$$

$$b = \overline{B} + \overline{C}. \overline{D} + C.D$$

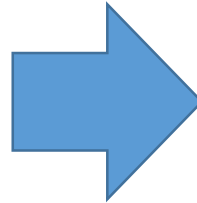
$$c = B + \overline{C} + D$$

$$d = \overline{B}. \overline{D} + C. \overline{D} + B. \overline{C}. D + A$$

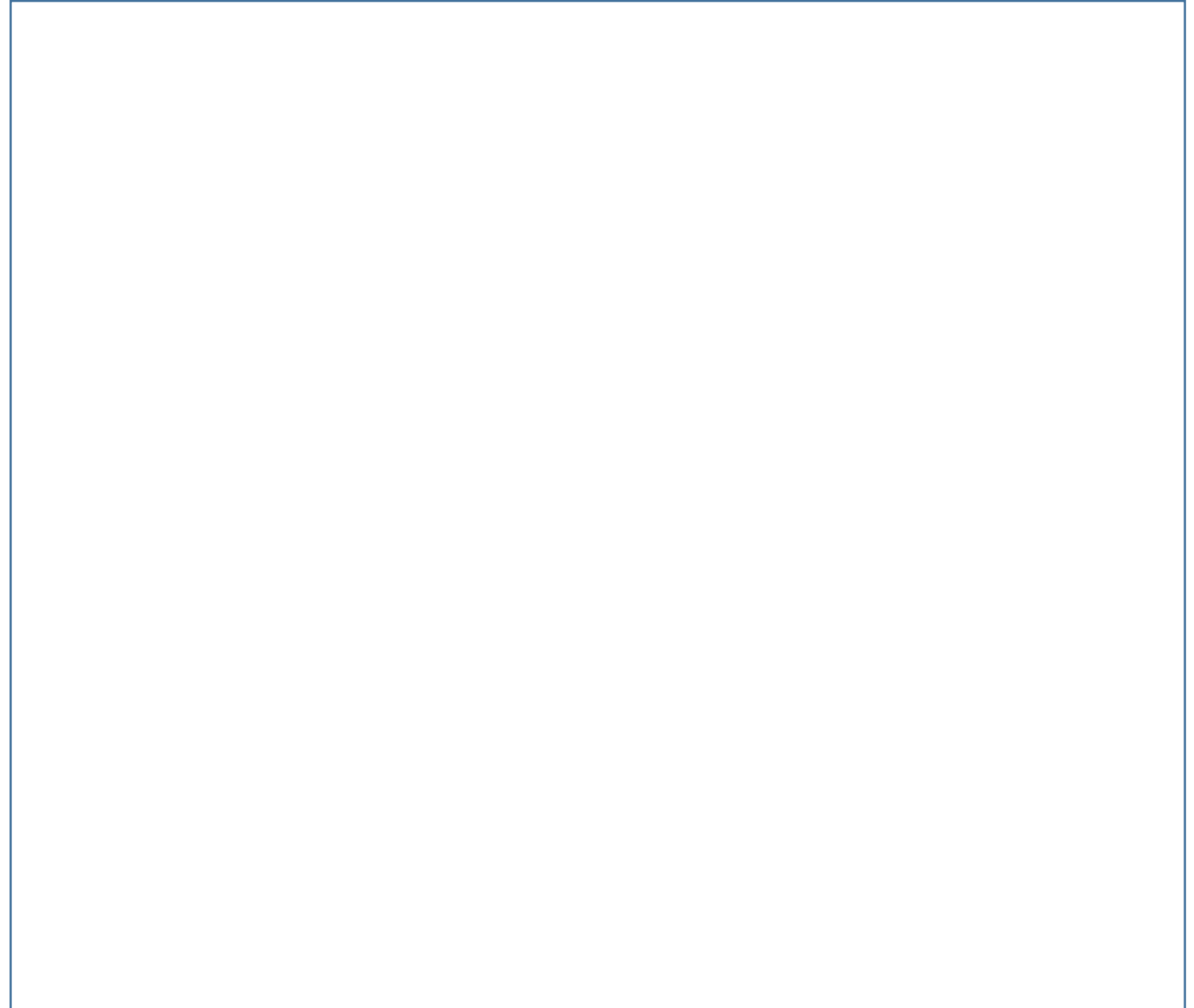
$$e = \overline{B}. \overline{D} + C. \overline{D}$$

$$f = A + \overline{C}. \overline{D} + B. \overline{C} + B. \overline{D}$$

$$g = A + B. \overline{C} + \overline{B}. C + C. \overline{D}$$



## Logic Circuit



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# Thanks!